
Q1. By BCC Proposition 3.3.4, the congruence

$$ax \equiv b \mod n$$

admits solutions if and only if $d = \gcd(a, n)$ divides b, and it admits exactly d non-congruent solutions modulo n. Since we seek a congruence with 11 non-congruent solutions, we set $d = 11$.

Noting that the prime factorization of 319 is $11 \cdot 29$, $\gcd(a, 319) = 11$ if $a \in \{11k \mid k \in \mathbb{Z} \text{ and } 29 \nmid k\}$. We choose $a = 11$ for ease.

Similarly, $11|b$ if $b \in \{11j \mid j \in \mathbb{N}\}$, so we can choose $b = 11$.

A congruence satisfying these requirements is

$$11x \equiv 11 \mod 319$$

$x_0 = 1$ is a solution to this congruence, and by Proposition 3.3.4, the set of pairwise non-congruent solutions is

$$\left\{x_0 + h\frac{319}{11} \mid h \in \{0, 1, \ldots, 10\}\right\} = \{1, 30, 59, 88, 117, 146, 175, 204, 233, 262, 291\}.$$

Q2. First, we can prove that g is a quadratic nonresidue mod p.

By Euler's Criterion,

$$\left(\frac{g}{p}\right) \equiv g^{\frac{p-1}{2}} \mod p$$

Squaring both sides and applying Fermat's little theorem gives

$$\left(\frac{g}{p}\right)^2 \equiv g^{p-1} \equiv 1 \pmod p$$

Therefore, the value of the Legendre symbol is restricted to $\left(\frac{g}{p}\right) = \pm 1$.

Since g is a generator of $(\mathbb{Z}/(p))^*$ its order is $(p-1)$, so $g^n \equiv 1 \mod p$ if and only if $(p-1) \mid n$.

Since $0 < \frac{p-1}{2} < p-1$, $\frac{p-1}{2}$ is clearly not a multiple of $p-1$, therefore

$$g^{\frac{p-1}{2}} \not\equiv 1 \mod p \implies \left(\frac{g}{p}\right) = -1.$$

By the multiplicative properties of the Legendre symbol, [BCC Proposition 5.2.22], we have that

$$\left(\frac{a}{p}\right) = \left(\frac{g^j}{p}\right) = \left(\frac{g}{p}\right)^j = (-1)^j$$

a is a quadratic residue mod p if and only if $\left(\frac{a}{p}\right) = \left(\frac{g^j}{p}\right) = 1$, and $(-1)^j = 1$ if and only if j is an even integer.

Therefore, $a = g^j$ is a quadratic residue mod p if and only if j is even, as required. □

Q3. First we prove that the three summations are equivalent.

$$\sum_{i=1}^{p}\left(\frac{i}{p}\right) = \sum_{i=1}^{p-1}\left(\frac{i}{p}\right) + \left(\frac{p}{p}\right) = \sum_{i=1}^{p-1}\left(\frac{i}{p}\right) + 0$$

Similarly,

$$\sum_{i=0}^{p-1}\left(\frac{i}{p}\right) = \left(\frac{0}{p}\right) + \sum_{i=1}^{p-1}\left(\frac{i}{p}\right) = 0 + \sum_{i=1}^{p-1}\left(\frac{i}{p}\right)$$

These both follow from the fact that

$$\left(\frac{a}{p}\right) = 0 \iff a \equiv 0 \mod p.$$

We use the observation from Question 2, and that

$$a \equiv b \mod p \implies \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$$

Since p is an (odd) prime, we can choose a generator $g \in (\mathbb{Z}/(p))^*$ such that any $a \in [1, \ldots, p-1]$ is congruent to g^j for some choice of $j \in [1, \ldots, p-1]$. We can rewrite the final summation as follows:

$$\sum_{i=1}^{p-1}\left(\frac{i}{p}\right) = \sum_{j=1}^{p-1}\left(\frac{g^j}{p}\right).$$

Let $m = \frac{p-1}{2}$. Then, we can split the summation into the odd and even exponents of g.

$$\begin{aligned}\sum_{j=1}^{p-1}\left(\frac{g^j}{p}\right) &= \sum_{k=1}^{m}\left(\frac{g^{2k-1}}{p}\right) + \sum_{k=1}^{m}\left(\frac{g^{2k}}{p}\right) \\ &= \sum_{k=1}^{m}(-1) + \sum_{k=1}^{m}(1) \\ &= -m + m = 0\end{aligned}$$

Therefore, we have

$$\sum_{i=1}^{p-1}\left(\frac{i}{p}\right) = \sum_{j=1}^{p-1}\left(\frac{g^j}{p}\right) = -m + m = 0$$

as required. □

Q4. Using the method outlined in ECC Theorem 1.5.13 for calculating infinite continued simple fractions, we take

$$\alpha_1 = \sqrt{11},\ a_1 = \lfloor \alpha_1 \rfloor = 3$$

Using the recursive definition $\alpha_{n+1} = 1/(\alpha_n - a_n)$, we calculate $\alpha_2, \alpha_3, \alpha_4$ to be

$$\begin{aligned}
\alpha_2 &= \frac{1}{\sqrt{11}-3} = \frac{\sqrt{11}+3}{2} && \implies a_2 = 3 \\
\alpha_3 &= \frac{1}{\frac{\sqrt{11}+3}{2}-3} = \frac{2}{\sqrt{11}-3} = \sqrt{11}+3 && \implies a_3 = 6 \\
\alpha_4 &= \frac{1}{\sqrt{11}+3-6} = \frac{1}{\sqrt{11}-3} = \alpha_2 && \implies a_4 = a_2 = 3
\end{aligned}$$

Since α_n only depends on α_{n-1}, it follows that $\alpha_n = \alpha_{n-2}$ for all $n \geq 4$.

Therefore, $[a_1; a_2, a_3, a_4, a_5, \dots] = [a_1; a_2, a_3, a_2, a_3, \dots]$, and $\sqrt{11} = [3; \overline{3, 6}]$ as required. $\square$